Lesson 8: Linking environment and history

90 minutes / introductory course / working draft

Reading evidence from tree rings

Observation means an inspected feature such as a ring width or scar. Dating places a ring in a calendar year through crossdating and anchoring. Interpretation concerns events such as felling, use, disturbance or climate change. A secure ring date does not automatically determine when timber was felled or used.

Archaeologists examine contexts and documents; ecologists compare trees and scars; climatologists calibrate and test against climate observations. Before combining sources, check differences in time, location and season.

Case: wooden containers and dated slips

A study of Nara containers compares ring dates with written dates on wooden slips. Distinguish the event recorded on a slip from the year of the last preserved ring. Examine association, preservation and the sequence of use and deposition.

The Nara National Research Institute for Cultural Properties publishes ring-width values, images and measurement paths. A paper and a public dataset do not necessarily contain the same specimens. Check specimen identifiers and archaeological context before linking them. Access routes appear at the end of this handout.

Exercise A: three timber dates

All examples are fictional. A document refers to a building in 1210. Samples were taken from different parts of the present structure. Spend 8 minutes on supported claims, limits and further evidence.

Timber A: last preserved ring 1187, bark absent. Timber B: last preserved ring 1199, bark present. Timber C: last preserved ring 1213, bark absent.

1. What does each year represent? What can be said about felling and use?
2. How could each timber relate to the 1210 record?
3. How could repair, reuse, storage, the meaning of the record and dating error be compared?
4. What should be checked about position, joints, working traces, bark and outer wood?

Hints

The number of outer rings lost from A and C is unknown, so the last preserved ring alone cannot determine the felling year. C formed a ring in 1213, prompting a reassessment of its relationship to the 1210 record. Later repair or additions are possibilities to evaluate using structure and context.

B formed a ring in 1199. Even with bark, inspect the outer ring and position within the ring before deciding felling season; the interval before use also remains to be considered. Timber dates alone do not fix a building's construction or repair date.

Exercise B: a research plan or literature review

Spend 8 minutes: draft for 5, review for 2 and revise for 1. Choose one format and state the question, sources, comparison and limits.

Research plan

Possible opening: “Were the timbers at a site associated with the building's construction or a later repair?” Specify the material, sampling and preservation, dating method and comparison evidence. What positions or working traces would be expected under a repair hypothesis? Compare another explanation. Identify alternatives if sampling is not possible.

This is an illustrative proposal, not a report of actual materials or findings. Do not present planned measurements or expected results as completed observations.

Literature review

Possible opening: “How can the relationship between temperature and ring width change over time?” Find relevant papers and compare region, species, season, calibration and verification periods, results and limits. Do not count articles based on the same data as independent evidence. Consider differences in samples and methods when results disagree.

Peer review

Do the sources address the question? Is there a competing explanation? What remains unresolved and what evidence could change the conclusion? Are figure sources, units and scope clear?

Report and routes to evidence

Choose a research plan or a research review. The report should be 2–4 A4 pages and cite at least three references; figures and tables may be included. Connect the question, materials and methods or literature selection, comparison, conclusion and limits. Check the course instructions for the submission method and deadline.

Finding papers and data

Follow a DOI to the publisher's article page and check the authors, year, title, methods and figures. Full text is not always freely available. Check a library or an author-hosted copy; if needed, politely request a copy from the corresponding author. For practical steps see Supplement BG06: accessing research papers.

- Containers and wooden slips: Maeda et al. (2024)
- Nara institute: archaeological tree-ring dataset announcement
- Hinoki and early spring temperature: Yonenobu and Eckstein (2006)
- Northeast Asia spring temperature: Ohyama et al. (2013)
- International Tree-Ring Data Bank: NOAA

Identify the source and scope of figures and data. Distinguish measurements, synthetic data and conceptual diagrams. Check the conditions for using external images and collaborative research materials.